

■ 3.2.10 Special Functions

<code>AiryAi[z]</code>	Airy function $\text{Ai}(z)$
<code>BesselJ[n, z]</code> and <code>BesselY[n, z]</code>	Bessel functions $J_n(z)$ and $Y_n(z)$
<code>BesselI[n, z]</code> and <code>BesselK[n, z]</code>	modified Bessel functions $I_n(z)$ and $K_n(z)$
<code>Beta[x, y]</code>	Euler beta function $B(x, y)$
<code>Beta[a, x, y]</code>	incomplete beta function $B_a(x, y)$
<code>Erf[z]</code>	error function $\text{erf}(z)$
<code>Erf[z0, z1]</code>	generalized error function $\text{erf}(z_1) - \text{erf}(z_0)$
<code>ExpIntegralE[n, z]</code>	exponential integral $E_n(z)$
<code>ExpIntegralEi[z]</code>	exponential integral $\text{Ei}(z)$
<code>Gamma[z]</code>	Euler gamma function $\Gamma(z)$
<code>Gamma[a, z]</code>	incomplete gamma function $\Gamma(a, z)$
<code>Gamma[a, z0, z1]</code>	generalized incomplete gamma function $\Gamma(a, z_1) - \Gamma(a, z_0)$
<code>Hypergeometric0F1[a, z]</code>	hypergeometric function ${}_0F_1(; a; z)$
<code>Hypergeometric1F1[a, b, z]</code>	Kummer confluent hypergeometric function ${}_1F_1(a; b; z)$
<code>HypergeometricU[a, b, z]</code>	confluent hypergeometric function $U(a, b, z)$
<code>Hypergeometric2F1[a, b, c, z]</code>	hypergeometric function ${}_2F_1(a, b; c; z)$
<code>LegendreP[n, z]</code> , <code>LegendreQ[n, z]</code>	Legendre functions $P_n(z)$ and $Q_n(z)$ of the first and second kinds
<code>LegendreP[n, m, z]</code> , <code>LegendreQ[n, m, z]</code>	associated Legendre functions $P_n^m(z)$ and $Q_n^m(z)$

Special functions, part one.

<code>LerchPhi[z, s, a]</code>	Lerch's transcendent $\Phi(z, s, a)$
<code>LogIntegral[z]</code>	logarithmic integral $\text{li}(z)$
<code>Pochhammer[a, n]</code>	Pochhammer symbol $(a)_n$
<code>PolyGamma[z]</code>	digamma function $\psi(z)$
<code>PolyGamma[n, z]</code>	n^{th} derivative of the digamma function $\psi^{(n)}(z)$
<code>PolyLog[n, z]</code>	polylogarithm function $\text{Li}_n(z)$
<code>Zeta[s]</code>	Riemann zeta function $\zeta(s)$
<code>Zeta[s, a]</code>	generalized Riemann zeta function $\zeta(s, a)$

Special functions, part two.

Mathematica includes all the common special functions of mathematical physics.

We will discuss each of the functions in turn. You can find more extensive discussions in various handbooks, such as *Handbook of Mathematical Functions*, edited by M. Abramowitz and I. Stegun, (Dover, 1965), *Formulas and Theorems for the Special Functions of Mathematical Physics*, by W. Magnus, F. Oberhettinger and R. P. Soni, (Third edition, Springer-Verlag, 1966), or *Encyclopedic Dictionary of Mathematics*, Volume IV, edited by Kiyosi Itô, (Second Edition, MIT Press, 1987).

There are often several conflicting definitions of any particular special function in the literature. When you use a special function in *Mathematica*, you should look at the definition given below to make sure it is what you want.

Mathematica gives exact results for some values of special functions.

```
In[1]:= Gamma[15/2]
Out[1]=  $\frac{135135 \sqrt{\pi}}{128}$ 
```

No exact result is known here.

```
In[2]:= Gamma[15/7]
Out[2]= Gamma[ $\frac{15}{7}$ ]
```

A numerical result, to arbitrary precision, can nevertheless be found.

```
In[3]:= N[%, 40]
Out[3]= 1.069071500448624397994137689702693267
```

You can give complex arguments to special functions.

```
In[4]:= Gamma[3 + 4I] //N
Out[4]= 0.00522554 - 0.172547 I
```

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Special functions automatically get applied to each element in a list.

```
In[5]:= Gamma[{3/2, 5/2, 7/2}]
Out[5]= { $\frac{\sqrt{\pi}}{2}$ ,  $\frac{3\sqrt{\pi}}{4}$ ,  $\frac{15\sqrt{\pi}}{8}$ }
```

Mathematica knows some analytical properties of special functions, such as derivatives.

```
In[6]:= D[Gamma[x], {x, 2}]
Out[6]= Gamma[x] PolyGamma[x]^2 + Gamma[x] PolyGamma[1, x]
```

You can use `FindRoot` to find roots of special functions.

```
In[7]:= FindRoot[BesselJ[0, x], {x, 1}]
Out[7]= {x -> 2.40483}
```

Special functions in *Mathematica* can usually be evaluated for arbitrary complex values of their arguments. Often, however, the defining relations given below apply only for some special choices of arguments. In these cases, the full function corresponds to a suitable extension or “analytic continuation” of these defining relations. Thus, for example, integral representations of functions are valid only when the integral exists, but the functions themselves can usually be defined elsewhere by analytic continuation.

As a simple example of how the domain of a function can be extended, consider the function represented by the sum $\sum_{k=0}^{\infty} x^k$. This sum converges only when $|x| < 1$. Nevertheless, it is easy to show analytically that for any x , the complete function is equal to $\frac{1}{1-x}$. Using this form, you can easily find a value of the function for any x , at least so long as $x \neq 1$.

Gamma and Related Functions

The **Euler gamma function** `Gamma[z]` is defined by the integral $\Gamma(z) = \int_0^{\infty} t^{z-1} e^{-t} dt$. For positive integer n , $\Gamma(n) = (n-1)!$. $\Gamma(z)$ can be viewed as a generalization of the factorial function, valid for complex arguments z .

The **Euler beta function** `Beta[x, y]` is $B(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)} = \int_0^1 t^{x-1}(1-t)^{y-1} dt$.

The **Pochhammer symbol** `Pochhammer[a, n]` is $(a)_n = \frac{\Gamma(a+n)}{\Gamma(a)}$. It often appears in series expansions for hypergeometric functions.

The **incomplete gamma function** `Gamma[a, z]` is defined by the integral $\Gamma(a, z) = \int_z^{\infty} t^{a-1} e^{-t} dt$. *Mathematica* includes a generalized incomplete gamma function `Gamma[a, z0, z1]` defined as $\int_{z_0}^{z_1} t^{a-1} e^{-t} dt$. The alternative incomplete gamma function $\gamma(a, z)$ can therefore be obtained in *Mathematica* as `Gamma[a, 0, z]`.

The **incomplete beta function** `Beta[a, x, y]` is given by $B_a(x, y) = \int_0^x t^{x-1}(1-t)^{y-1} dt$. Notice that the limit of the integral appears as

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the *first* argument of the incomplete beta function, and the *second* argument of the incomplete gamma functions.

The incomplete beta and gamma functions, and their inverses, are common in statistics.

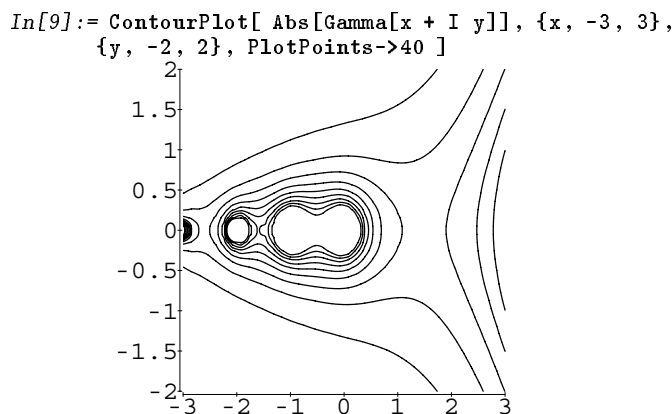
Derivatives of the gamma function often appear in summing rational series. The **digamma function** `PolyGamma[z]` is the logarithmic derivative of the gamma function, given by $\psi(z) = \frac{\Gamma'(z)}{\Gamma(z)}$. For integer arguments, the digamma function satisfies the relation $\psi(n) = -\gamma + \sum_{k=1}^{n-1} \frac{1}{k}$, where γ is Euler's constant (`EulerGamma` in *Mathematica*).

The **polygamma functions** `PolyGamma[n, z]` are given by $\psi^{(n)}(z) = \frac{d^n}{dz^n} \psi(z)$. Notice that the digamma function corresponds to $\psi^{(0)}(z)$: $\psi^{(n)}(z)$ is the $(n+1)^{\text{th}}$, not the n^{th} , logarithmic derivative of the gamma function. The polygamma functions satisfy the relation $\psi^{(n)}(z) = (-1)^{n+1} n! \sum_{k=0}^{\infty} \frac{1}{(z+k)^{n+1}}$.

Many exact results for gamma and polygamma functions are built in to *Mathematica*.

```
In[8]:= PolyGamma[6]
Out[8]=  $\frac{137}{60} - \text{EulerGamma}$ 
```

Here is a contour plot of the gamma function in the complex plane. The poles in the function at negative integers are apparent.



Zeta and Related Functions

The **Riemann zeta function** `Zeta[s]` is defined by the relation $\zeta(s) = \sum_{k=1}^{\infty} k^{-s}$ (for $s > 1$). Zeta functions with integer arguments arise in evaluating various sums and integrals. *Mathematica* gives exact results when possible for zeta functions with integer arguments.

There is an analytic continuation of $\zeta(s)$ for arbitrary complex $s \neq 1$. The

zeta function for complex argument is central to number theoretical studies of the distribution of primes. Of particular importance are the values on the critical line $\text{Re } s = \frac{1}{2}$.

The **generalized Riemann zeta function** or **Hurwitz zeta function** $\text{Zeta}[s, a]$ is given by $\zeta(s, a) = \sum_{k=0}^{\infty} (k+a)^{-s}$, where any term with $k+a=0$ is excluded.

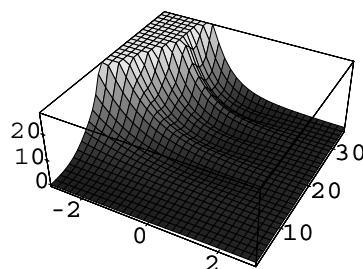
Mathematica gives exact results for $\zeta(2n)$.

```
In[10]:= Zeta[6]
```

```
Out[10]=  $\frac{\pi^6}{945}$ 
```

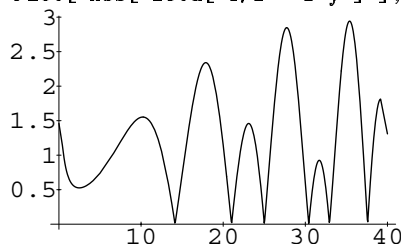
Here is a three-dimensional picture of the Riemann zeta function in the complex plane. The cover of this book was made from a different view of the same picture.

```
In[11]:= Plot3D[Abs[Zeta[x + I y]], {x, -3, 3}, {y, 2, 35}, PlotPoints->30]
```



This is a plot of the absolute value of the Riemann zeta function on the critical line $\text{Re}(z) = \frac{1}{2}$. You can see the first few zeroes of the zeta function.

```
In[12]:= Plot[Abs[Zeta[1/2 + I y]], {y, 0, 40}]
```



The **polylogarithm functions** $\text{PolyLog}[n, z]$ are given by $\text{Li}_n(z) = \sum_{k=1}^{\infty} \frac{z^k}{k^n}$. The **dilogarithm** $\text{PolyLog}[2, z]$ satisfies $\text{Li}_2(z) = \int_z^0 \frac{\log(1-t)}{t} dt$. $\text{Li}_2(1-z)$ is sometimes known as **Spence's integral**. Polylogarithms crop up in Feynman diagram integrals in elementary particle physics. The polylogarithm function is sometimes known as **Jonquière's function**.

The **Lerch transcendent** $\text{LerchPhi}[z, s, a]$ is a generalization of the zeta and polylogarithm functions, given by $\Phi(z, s, a) = \sum_{k=0}^{\infty} \frac{z^k}{(a+k)^s}$, where any term with

$k + a = 0$ is excluded. Many sums of reciprocal powers can be expressed in terms of the Lerch transcendent. For example, the **Catalan beta function** $\beta(s) = \sum_{k=0}^{\infty} (-1)^k (2k+1)^{-s}$ can be obtained as $2^{-s} \Phi(-1, s, \frac{1}{2})$.

The Lerch transcendent is related to integrals of the **Fermi-Dirac** distribution in statistical mechanics by $\int_0^{\infty} \frac{k^n}{e^{k-\mu} + 1} dk = e^{\mu} \Phi(e^{\mu}, n+1, 1)$.

The Lerch transcendent can also be used to evaluate **Dirichlet L series** which appear in number theory. The basic L series has the form $L(s, \chi) = \sum_{k=1}^{\infty} \chi(k) k^{-s}$, where the “character” $\chi(k)$ is an integer function with period m . L series of this kind can be written as sums of Lerch functions with z a power of $e^{2\pi i/m}$.

`LerchPhi[z, s, a, DoublyInfinite->True]` gives the doubly-infinite sum $\sum_{k=-\infty}^{\infty} \frac{z^k}{(a+k)^s}$.

Exponential Integral and Related Functions

Mathematica has two forms of exponential integral: `ExpIntegralE` and `ExpIntegralEi`.

The **exponential integral function** `ExpIntegralE[n, z]` is defined by $E_n = \int_1^{\infty} \frac{e^{-zt}}{t^n} dt$.

The second **exponential integral function** `ExpIntegralEi[z]` is defined by $Ei(z) = -\int_{-z}^{\infty} \frac{e^{-t}}{t} dt$, where the principal value of the integral is taken.

The **logarithmic integral function** `LogIntegral[z]` is given by $li(z) = \int_0^z \frac{dt}{\log t}$, where the principal value of the integral is taken. $li(z)$ is central to the study of the distribution of primes in number theory. The logarithmic integral function is sometimes also denoted by $Li(z)$. In some number-theoretical applications, $li(z)$ is defined as $\int_2^z \frac{dt}{\log t}$, with no principal value taken. This differs from the definition used in *Mathematica* by the constant $li(2)$.

The **sine and cosine integral functions** $Si(z) = \int_0^z \frac{\sin(t)}{t} dt$ and $Ci(z) = -\int_z^{\infty} \frac{\cos(t)}{t} dt$ are given in terms of the exponential integral function by $Si(z) = \frac{1}{2i}[E_1(iz) - E_1(-iz)] + \frac{\pi}{2}$ and $Ci(z) = -\frac{1}{2}[E_1(iz) + E_1(-iz)]$.

Error Function

The **error function** $\text{Erf}[z]$ is the integral of the Gaussian distribution, given by $\text{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-t^2} dt$.

The generalized error function $\text{Erf}[z_0, z_1]$ is defined by the integral $\frac{2}{\sqrt{\pi}} \int_{z_0}^{z_1} e^{-t^2} dt$. The **complementary error function** $\text{erfc}(z)$ can be obtained as $\text{Erf}[z, \text{Infinity}]$.

The error function is central to many calculations in statistics.

Closely related to the error function are the **Fresnel integrals**, $C(z) = \int_0^z \cos\left(\frac{\pi t^2}{2}\right) dt$ and $S(z) = \int_0^z \sin\left(\frac{\pi t^2}{2}\right) dt$. These are given in terms of the error function by $C(z) + iS(z) = \frac{1+i}{2} \text{erf}\left(\frac{\sqrt{\pi}}{2}(1-i)z\right)$. Fresnel integrals occur in diffraction theory.

Bessel Functions

The **Bessel functions** $\text{BesselJ}[n, z]$ and $\text{BesselY}[n, z]$ are linearly independent solutions to the differential equation $z^2 \frac{d^2 y}{dz^2} + z \frac{dy}{dz} + (z^2 - n^2)y = 0$. For integer n , the $J_n(z)$ are regular at $z = 0$, while the $Y_n(z)$ have a logarithmic divergence at $z = 0$.

Bessel functions arise in solving differential equations for systems with cylindrical symmetry.

$J_n(z)$ is often called the **Bessel function of the first kind**, or simply *the* Bessel function. $Y_n(z)$ is referred to as the **Bessel function of the second kind**, the **Weber function**, or the **Neumann function** (denoted $N_n(z)$).

The **Hankel functions** (or **Bessel functions of the third kind**) $H_n^{(1,2)}(z) = J_n(z) \pm iY_n(z)$ give an alternative pair of solutions to the Bessel differential equation.

In studying systems with spherical symmetry, **spherical Bessel functions** arise, defined by $f_n(z) = \sqrt{\frac{\pi}{2z}} F_{n+\frac{1}{2}}(z)$, where f, F can be j, J, y, Y , or h^i, H^i . For integer n , *Mathematica* gives exact algebraic formulae for spherical Bessel functions.

The **modified Bessel functions** $\text{BesselI}[n, z]$ and $\text{BesselK}[n, z]$ are solutions to the differential equation $z^2 \frac{d^2 y}{dz^2} + z \frac{dy}{dz} - (z^2 + n^2)y = 0$. For integer n , $I_n(z)$ is regular at $z = 0$; $K_n(z)$ always has a logarithmic divergence at $z = 0$. The $I_n(z)$ are sometimes known as **hyperbolic Bessel functions**.

Particularly in electrical engineering, one often defines the **Kelvin functions**, according to $\text{ber}_n(z) + i \text{bei}_n(z) = e^{n\pi i} J_n(ze^{-\pi i/4})$, $\text{ker}_n(z) + i \text{kei}_n(z) = e^{-n\pi i/2} K_n(ze^{\pi i/4})$.

The **Airy function** $\text{AiryAi}[z]$ is a solution to the differential equation $\frac{d^2 y}{dx^2} - xy = 0$.

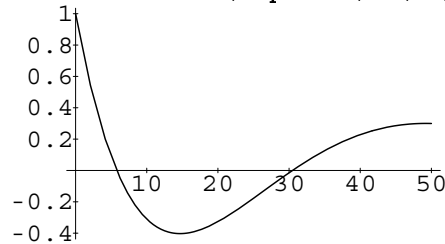
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It is related to Bessel functions with one-third-integer orders. $\text{Ai}(z)$ appears in electromagnetic theory and quantum mechanics.

Here is a plot of $J_0(\sqrt{x})$. This is a curve that an idealized chain hanging from one end can form when you wiggle it.

```
In[13]:= Plot[ BesselJ[0, Sqrt[x]], {x, 0, 50} ]
```



Mathematica generates explicit formulae for half-integer-order Bessel functions.

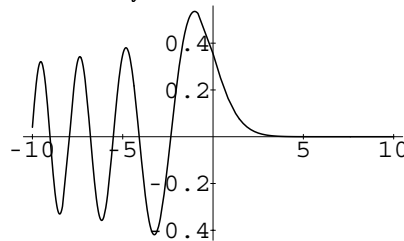
```
In[14]:= BesselK[3/2, x]
```

```
Out[14]= 
$$\frac{\sqrt{\pi} \left(1 + \frac{1}{x}\right)}{\sqrt{2} e^x \sqrt{x}}$$

```

The Airy function plotted here gives the quantum-mechanical amplitude for a particle in a potential that increases linearly from left to right. The amplitude is exponentially damped in the classically inaccessible region on the right.

```
In[15]:= Plot[ AiryAi[x], {x, -10, 10} ]
```



Legendre and Related Functions

The **Legendre functions** and **associated Legendre functions** satisfy the differential equation $(1 - z^2)\frac{d^2y}{dz^2} - 2z\frac{dy}{dz} + [(n(n+1) - \frac{m^2}{1-z^2})]y = 0$. The Legendre functions of the first kind, $\text{LegendreP}[n, z]$ and $\text{LegendreP}[n, m, z]$, reduce to Legendre polynomials when n and m are integers. The **Legendre functions of the second kind** $\text{LegendreQ}[n, z]$ and $\text{LegendreQ}[n, m, z]$ give the second linearly independent solution to the differential equation. They have logarithmic singularities at $z = \pm 1$. The $P_n(z)$ and $Q_n(z)$ solve the differential equation with $m = 0$.

In general, both $P_n^m(z)$ and $Q_n^m(z)$ have a branch cut along the real axis running from $-\infty$ to $+1$. For integer n and even m , however, $P_n^m(z)$ is a Legendre polynomial, and has a unique value for all z .

It is often necessary to find values for Legendre functions with z a real num-

ber in the range $-1 < z < +1$. In this case, the different values from the two sides of the branch cut are conventionally combined to give a unique result using the formulae $P_n^m(x) = [e^{im\pi/2}P_n^m(x+i0) + e^{-im\pi/2}P_n^m(x-i0)]/2$ and $Q_n^m(x) = e^{-im\pi}[e^{-im\pi/2}Q_n^m(x+i0) + e^{im\pi/2}Q_n^m(x-i0)]/2$.

The two versions of Legendre functions are distinguished by settings for the option `LegendreType`. `LegendreType->Complex` gives Legendre functions with a branch cut from $-\infty$ to $+1$. `LegendreType->Real` combines values from the two sides of the branch cut to give functions with branch cuts from $-\infty$ to -1 , and $+1$ to $+\infty$. These functions are continuous on the interval $(-1, 1)$.

The Legendre functions crop up in studies of quantum mechanical scattering processes.

Toroidal functions or **ring functions**, which arise in studying systems with toroidal symmetry, can be expressed in terms of the Legendre functions $P_{\nu-\frac{1}{2}}^{\mu}(\cosh \eta)$ and $Q_{\nu-\frac{1}{2}}^{\mu}(\cosh \eta)$.

Conical functions can be expressed in terms of $P_{-\frac{1}{2}+ip}^{\mu}(\cos \theta)$ and $Q_{-\frac{1}{2}+ip}^{\mu}(\cos \theta)$.

When you use the function `LegendreP[n, x]` with an integer n , you get a Legendre polynomial. If you take n to be an arbitrary complex number, you get in general a Legendre function.

In the same way, you can use the functions `GegenbauerC` and so on with arbitrary complex indices to get **Gegenbauer functions**, **Chebyshev functions**, **Hermite functions**, **Jacobi functions** and **Laguerre functions**.

Confluent Hypergeometric Functions

Many of the special functions that we have discussed so far can be viewed as special cases of the **confluent hypergeometric function** `Hypergeometric1F1[a, b, z]`.

The confluent hypergeometric function can be obtained from the series expansion ${}_1F_1(a; b; z) = 1 + \frac{az}{b} + \frac{a(a+1)}{b(b+1)} \frac{z^2}{2!} + \dots = \sum_{k=0}^{\infty} \frac{(a)_k}{(b)_k} \frac{z^k}{k!}$. Some special results are obtained when a and b are both integers. If $a < 0$, and either $b > 0$ or $b < a$, the series yields a polynomial with a finite number of terms. If integer $b \leq 0$, ${}_1F_1(a; b; z)$ is undefined.

Among the functions that can be obtained from ${}_1F_1$ are the Bessel functions, error function, incomplete gamma function, and Hermite and Laguerre polynomials.

The function ${}_1F_1(a; b; z)$ is sometimes denoted $\Phi(a; b; z)$ or $M(a, b, z)$. It is often known as the **Kummer function**.

The ${}_1F_1$ function can be written in the integral representation

$${}_1F_1(a; b; z) = \frac{\Gamma(b)}{\Gamma(b-a)\Gamma(a)} \int_0^1 e^{zt} t^{a-1} (1-t)^{b-a-1} dt.$$

The ${}_1F_1$ confluent hypergeometric function is a solution to Kummer's differential equation $z \frac{d^2 y}{dz^2} + (b-z) \frac{dy}{dz} - ay = 0$, with the boundary conditions ${}_1F_1(a; b; 0) = 1$ and $\frac{\partial}{\partial z} {}_1F_1(a; b; z)|_{z=0} = \frac{a}{b}$.

The function `HypergeometricU[a, b, z]` gives a second linearly independent solution to Kummer's equation. This function behaves like z^{1-b} for small z . It has a branch cut along the negative real axis in the complex z plane.

The function $U(a, b, z)$ has the integral representation

$$U(a, b, z) = \frac{1}{\Gamma(a)} \int_0^\infty e^{-zt} t^{a-1} (1+t)^{b-a-1} dt.$$

Special cases of U give such functions as the exponential and logarithmic integral functions, and modified Bessel functions.

$U(a, b, z)$, like ${}_1F_1(a; b; z)$, is sometimes known as the **Kummer function**, and sometimes denoted by Ψ .

The **Whittaker functions** give an alternative pair of solutions to Kummer's differential equation. The Whittaker function $M_{\kappa, \mu}$ is related to ${}_1F_1$ by $M_{\kappa, \mu}(z) = e^{-z/2} z^{\frac{1}{2}+\mu} {}_1F_1(\frac{1}{2} + \mu - \kappa; 1 + 2\mu; z)$. The second Whittaker function $W_{\kappa, \mu}$ obeys the same relation, with ${}_1F_1$ replaced by U .

The **parabolic cylinder functions** are related to Whittaker functions by $D_\nu(z) = 2^{\frac{1}{4}+\frac{\nu}{2}} z^{-\frac{1}{2}} W_{\frac{1}{4}+\frac{\nu}{2}, -\frac{1}{4}}(\frac{z^2}{2})$. For integer ν , the parabolic cylinder functions reduce to Hermite polynomials.

The **Coulomb wave functions** are also special cases of the confluent hypergeometric function. Coulomb wave functions give solutions to the radial Schrödinger equation in the Coulomb potential of a point nucleus. The regular Coulomb wave function is given by $F_L(\eta, \rho) = C_L(\eta) \rho^{L+1} e^{-i\rho} {}_1F_1(L+1-i\eta; 2L+2; 2i\rho)$, where $C_L(\eta) = [2^L e^{-\pi\eta/2} |\Gamma(L+1+i\eta)|] / \Gamma(2L+2)$.

Other special cases of the confluent hypergeometric function include the **Toronto functions** $T(m, n, r)$, **Poisson-Charlier polynomials** $\rho_n(\nu, x)$, **Cunningham functions** $\omega_{n,m}(x)$ and **Bateman functions** $k_\nu(x)$.

A limiting form of the confluent hypergeometric function which often appears is `Hypergeometric0F1[a, z]`. This function is obtained as the limit ${}_0F_1(; a; z) = \lim_{q \rightarrow \infty} {}_1F_1(q; a; \frac{z}{q})$.

The ${}_0F_1$ function has the series expansion ${}_0F_1(; a; z) = \sum_{k=0}^\infty \frac{z^k}{(a)_k k!}$ and satisfies the differential equation $z \frac{d^2 y}{dz^2} + a \frac{dy}{dz} - y = 0$.

Bessel functions of the first kind can be expressed in terms of the ${}_0F_1$ function.

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Hypergeometric Functions

The **hypergeometric function** `Hypergeometric2F1[a, b, c, z]` has series expansion ${}_2F_1(a, b; c; z) = \sum_{k=0}^{\infty} \frac{(a)_k (b)_k}{(c)_k} \frac{z^k}{k!}$. The function is a solution of the hypergeometric differential equation $z(1-z)\frac{d^2y}{dz^2} + [c - (a+b+1)z]\frac{dy}{dz} - aby = 0$.

The hypergeometric function can also be written as an integral: ${}_2F_1(a, b; c; z) = \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \int_0^1 t^{b-1} (1-t)^{c-b-1} (1-tz)^{-a} dt$.

The hypergeometric function is also sometimes denoted by F , and is known as the **Gauss series** or the **Kummer series**.

The Legendre functions, and the functions which give generalizations of other orthogonal polynomials, can be expressed in terms of the hypergeometric function. Complete elliptic integrals can also be expressed in terms of the ${}_2F_1$ function.

The **Riemann P function**, which gives solutions to Riemann's differential equation, is also a ${}_2F_1$ function.